

Ricketts Analysis

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Robert Ricketts believed that cephalometric analysis was one of the most valuable tools available for diagnosing and monitoring patients, as well as for evaluating their growth and development. In 1969, he developed a computerized analysis intended for routine use by clinicians using a lateral and frontal cephalometric tracing and a long-range growth projection to maturity.

Figure 8-1 shows the Ricketts lateral cephalometric tracing. Figures 8-2 to 8-46 identify and describe the craniofacial landmarks and planes used in the Ricketts analysis.

Interpretation

Chin in space

Facial axis

The angle formed between the Ba-N plane and the plane from foramen rotundum (PT) to Gn. On average, this angle is 90 degrees. A lesser angle suggests a retropositioned

chin, whereas an angle greater than 90 degrees suggests a protrusive or forward-growing chin.

Facial (depth) angle

The angle between the facial plane (N-Pog) and FH. This angle provides some indication of the horizontal position of the chin. It also suggests whether a skeletal Class II or III pattern is caused by the position of the mandible.

Mandibular plane

Measures an angle to FH. On average, this angle is 26 degrees at 9 years of age and decreases approximately 1 degree every 3 years. A high or steep mandibular plane angle implies that an open bite may be caused by the skeletal morphologic characteristics of the mandible. A low mandibular plane suggests the opposite (ie, a deep bite).

Convexity

Convexity at point A

The convexity of the middle face is measured from point A to the facial plane (N-Pog). The clinical norm at 9 years of

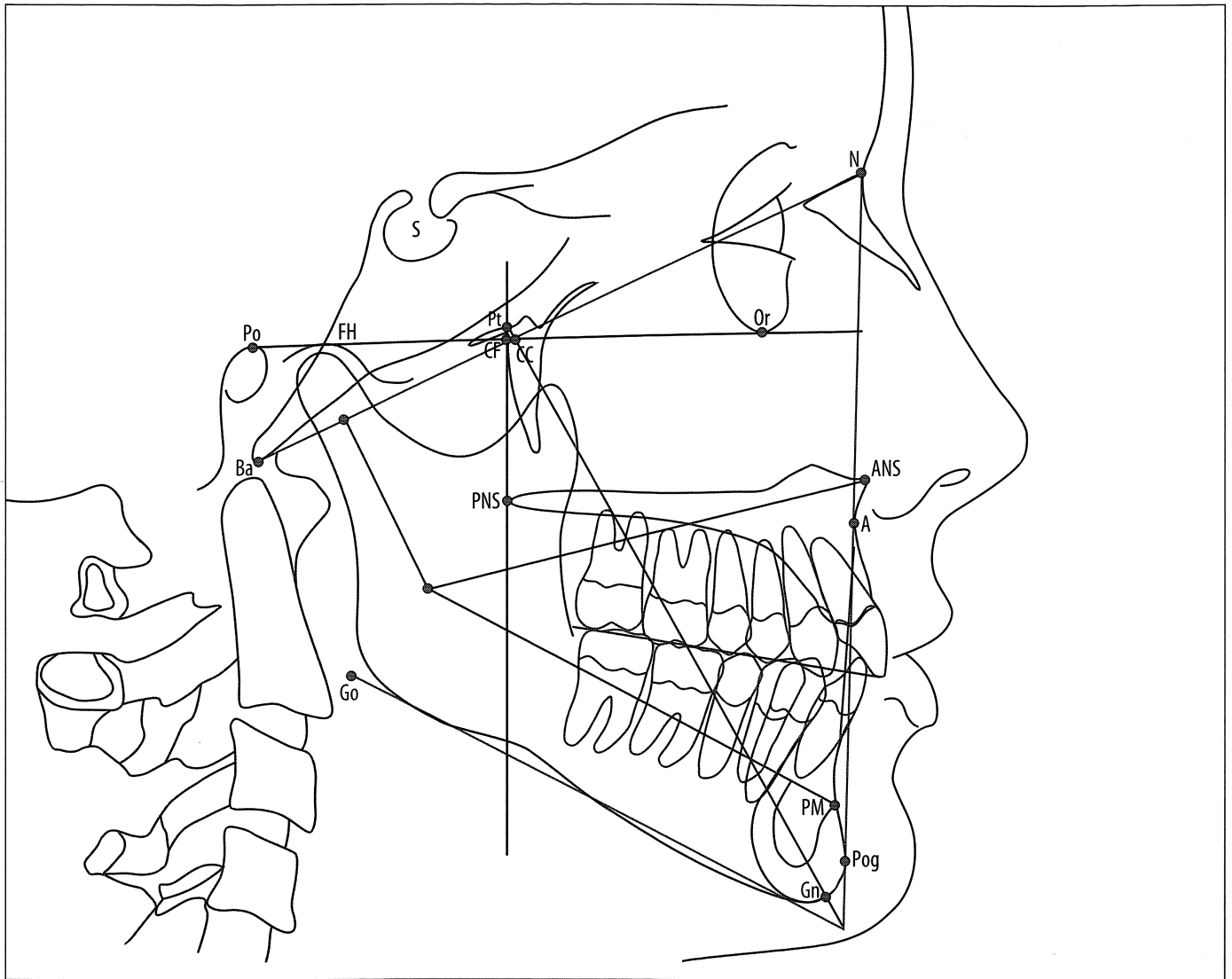


Fig 8-1 Lateral cephalometric tracing of a woman with good facial balance and proportion and an ideal Class I occlusion. The following anatomic landmarks and cephalometric planes are shown.

N (nasion): The point in the midline of both the nasal root and the nasofrontal suture.

Or (orbitale): The lowest point on the lower margin of each orbit.

Po (porion): The highest point on the upper margin of the external cutaneous auditory meatus.

FH (Frankfort horizontal plane): The line connecting Po and Or.

Ba (basion): The most inferior posterior point of the occipital bone at the anterior margin of the occipital foramen.

Pt (point): The intersection of the inferior border of the foramen rotundum with the posterior wall of the pterygomaxillary fissure.

CC (center of cranium) point: Cephalometric landmark formed by the intersection of the two lines Ba-N and Pt-Gn.

CF (center of face) point: Cephalometric landmark formed by the intersection of FH and the perpendicular through Pt point.

ANS (anterior nasal spine).

PNS (posterior nasal spine).

Point A: The deepest point in the curve of the maxilla between the ANS and the dental alveolus.

PM (protuberance menti or suprapogonion).

Pog (pogonion): The most anterior midpoint of the mandible.

Gn (gnathion): The lowest point in the midline on the lower border of the chin.

Go (gonion): The most lateral point on the mandibular angle close to the bony gonion.

Fig 8-2 Xi point: A point located at the geometric center of the ramus. Location of Xi is keyed geometrically to Po-Or (FH) and perpendicular through Pt (pterygoid vertical [PtV]; a line perpendicular to FH at the posterior margin of the pterygopalatine fossa) in the following steps:

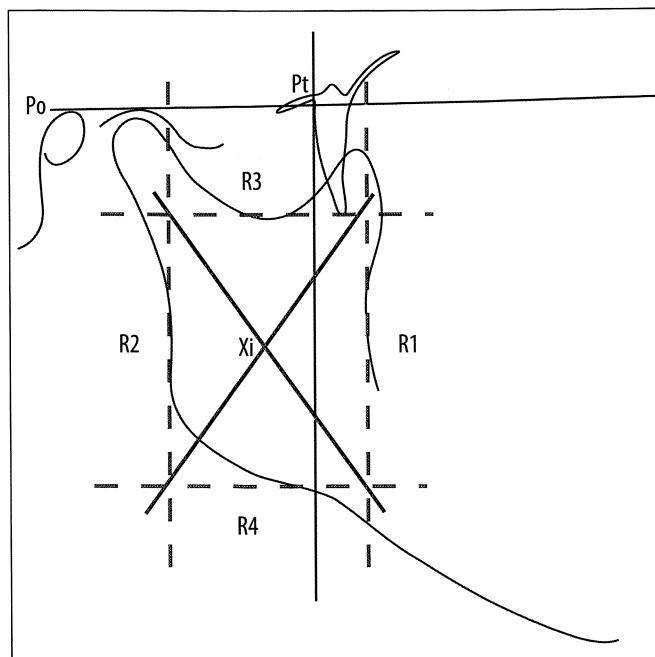
1. Planes perpendicular to FH and PtV are constructed.
2. The constructed planes are tangent to points R1, R2, R3, and R4 on the borders of the ramus.
3. The constructed planes form a rectangle enclosing the ramus.
4. Xi is located in the center of the rectangle at the intersection of the diagonals.

R1-mandible: The deepest point on the curve of the anterior border of the ramus, one half the distance between the inferior and superior curves.

R2-mandible: A point located on the posterior border of the ramus of the mandible.

R3-mandible: A point located at the center and most inferior aspect of the sigmoid notch of the ramus of the mandible.

R4-mandible: A point on the lower border of the mandible, directly inferior to the center of the sigmoid notch of the ramus.



age is 2 mm and decreases 1 degree every 5 years. High convexity suggests a Class II skeletal pattern; negative convexity suggests a Class III skeletal pattern.

Teeth

Mandibular incisor to A-Pog

The A-Pog line or plane is referred to as the *dental plane* and is a useful reference line from which to measure the position of the anterior teeth. Ideally, the mandibular incisor should be located 1 mm ahead of the A-Pog line. This measurement is used to define the protrusion of the mandibular arch.

Maxillary molar to PtV

This measurement is the distance from PtV (back of the maxilla) to the distal of the maxillary molar. On average, this measurement should equal the age of the patient plus 3 mm (eg, a patient 11 years of age has a norm of $11 + 3 = 14$

mm). This measurement assists in determining whether the malocclusion is due to the position of the maxillary or mandibular molar. It is also useful in deciding whether extractions are necessary.

Mandibular incisor inclinations (1 to A-Pog)

The angle between the long axis of the mandibular incisor and the A-Pog plane (1 to A-Pog) is measured to provide some idea of mandibular incisor procumbency. On the average, this angle should be 22 ± 4 degrees.

Profile

Lower lip to E-line

The distance between the lower lip and the esthetic (nose-chin) line is an indication of the soft tissue balance between the lips and the profile. The average norm for this measurement is -2 mm at 9 years of age. The positive values are those ahead of the E-line.

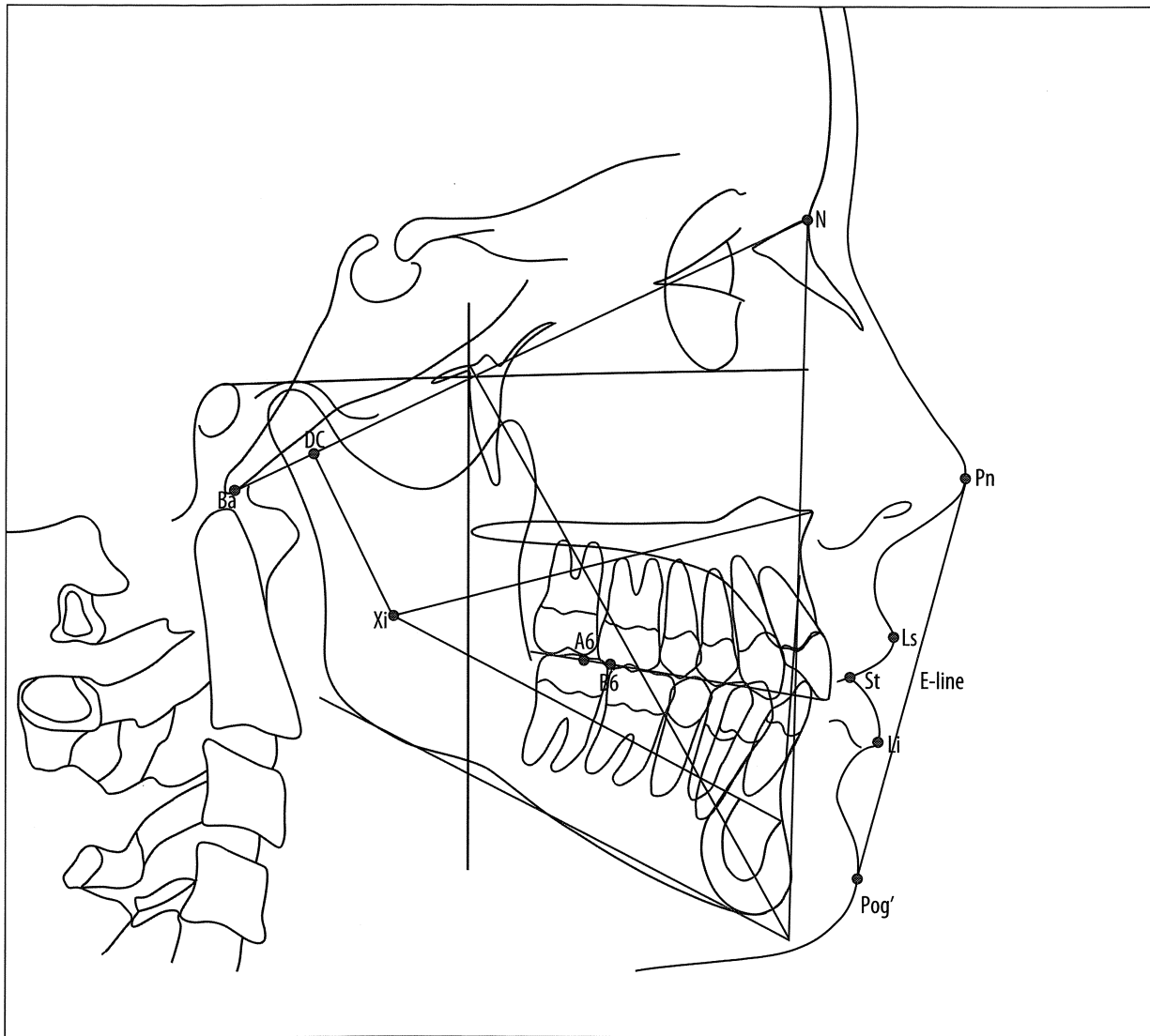


Fig 8-3 *B6 (mandibular molar)*: A point on the occlusal plane perpendicular to the distal surface of the crown of the mandibular first molar.

A6 (maxillary molar): A point on the occlusal plane perpendicular to the distal surface of the crown of the maxillary first molar.

DC point: Cephalometric landmark representing the center of the neck of the condyle on the Ba-N line.

X_i : The geometric center of the ramus.

Pn (pronasale): The most protruded point of the nasal tip.

Pog' (soft tissue pogonion): The most protruding point of the soft tissue chin contour.

St (stomion): The midpoint of the labial fissure between gently closed lips.

Ls (Labrale superior): Most anterior point of the upper lip.

Li (Labrale inferior): Most anterior point on the lower lip.

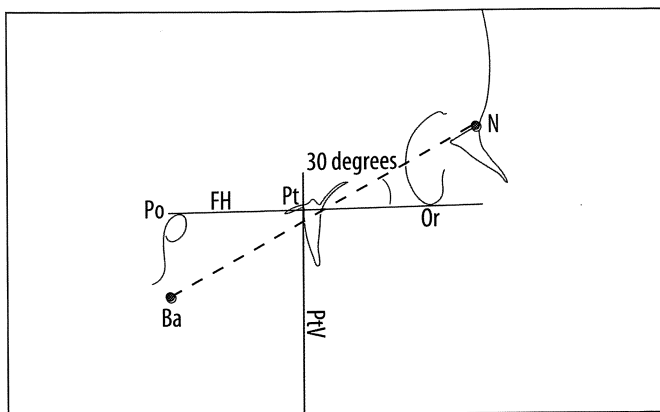


Fig 8-4 Two lateral cephalometric reference planes are used: the Frankfort horizontal (FH), a line drawn through Po and Or; and the pterygoid vertical (PtV), a line perpendicular to FH through Pt. From there, the other planes can be constructed. *Cranial base plane*: Constructed by a line drawn from N to Ba.

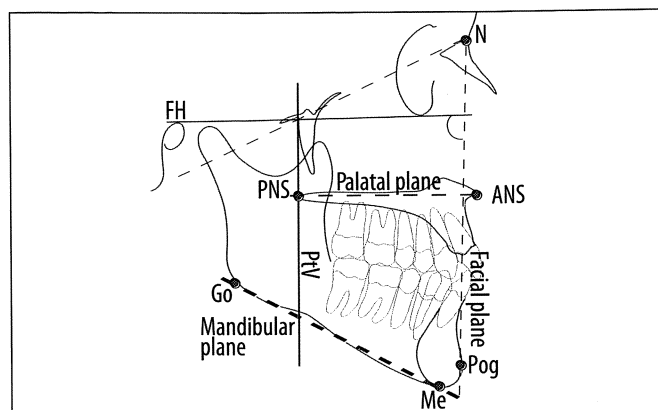


Fig 8-5 *Facial plane*: Constructed by a line drawn from N to Pog. This represents the anteroposterior position of the mandible measured relative to FH. *Palatal plane*: Constructed by a line drawn from ANS to PNS. *Mandibular plane*: Constructed by a line drawn from Go to menton (Me) tangent to the inferior border of the mandible.

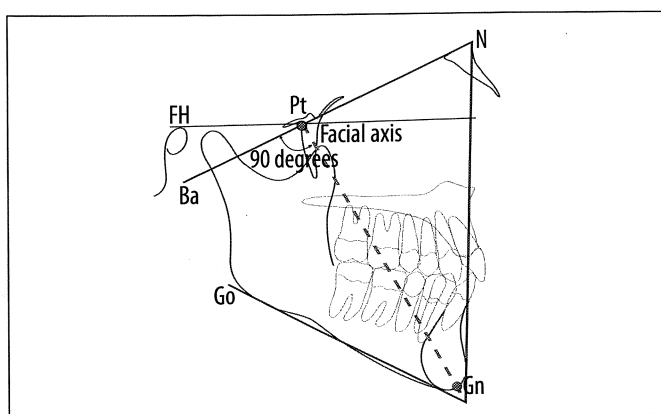


Fig 8-6 *Facial axis*: A line drawn from Pt through Gn.

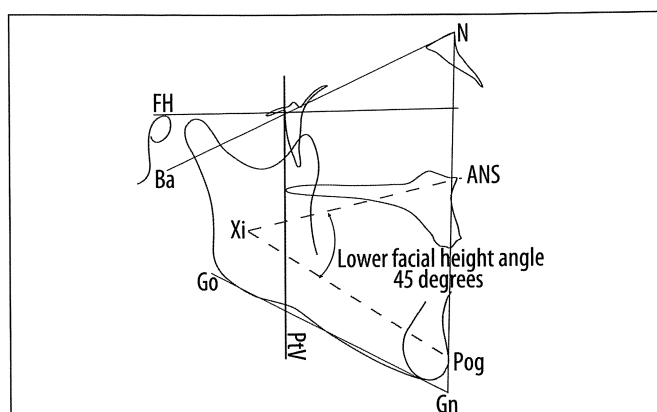


Fig 8-7 *Lower facial height*: The intersection of two planes: ANS-Xi and Xi-Pog.

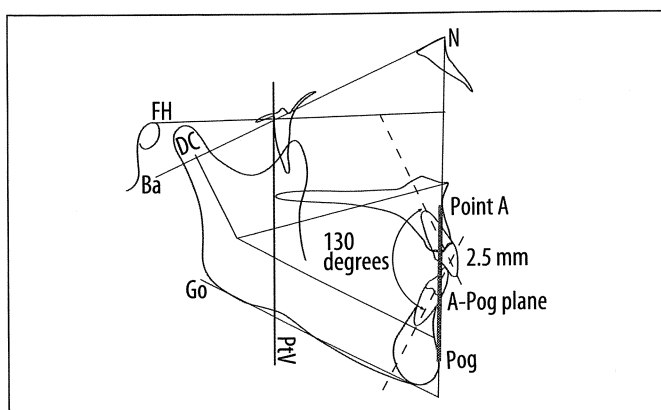


Fig 8-8 *A-Pog plane (dental plane)*: A reference plane for determining mandibular incisor protrusion, as indicated by the linear distance between the A-Pog plane and the tip of the mandibular incisor.

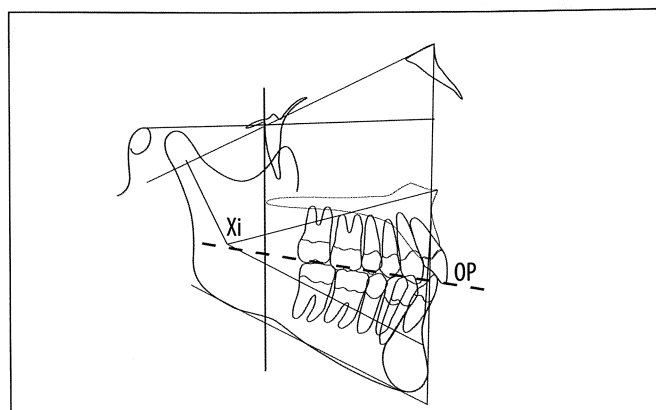


Fig 8-9 *Occlusal plane (OP)*: A line bisecting the cusp tips of the molars and passing through the cusp tips of the first premolars. In the white adult, the plane passes just inferior to Xi, nearly bisecting the angle of lower facial height. The long axis of the maxillary incisor intersects the long axis of the mandibular incisor at an angle of approximately 130 degrees, with a 2.5-mm horizontal (overjet) relationship and a 2.5-mm vertical (overbite) relationship.

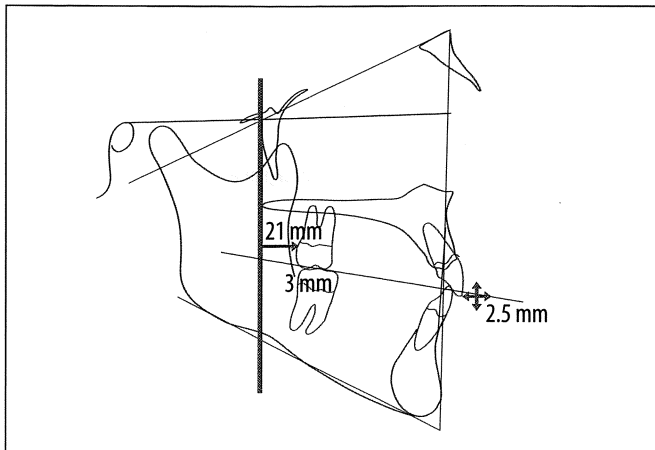


Fig 8-10 Molar position: The maxillary first molar is measured from PtV to the distal of the maxillary first molar. Normal = 21 mm. The mandibular molar is 3 mm anterior to the maxillary molar.

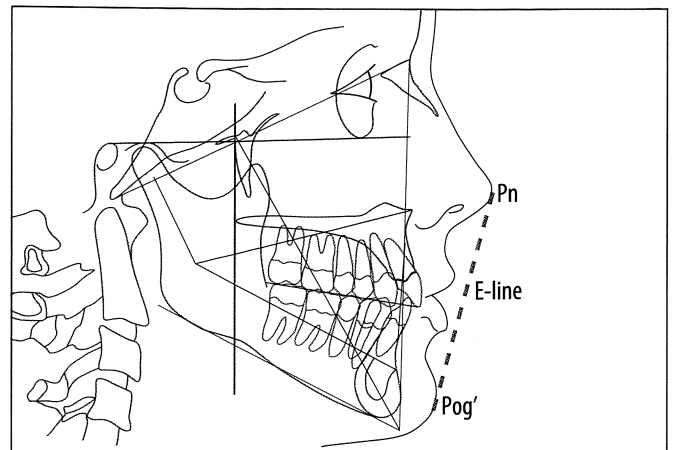


Fig 8-11 Soft tissue profile: A line connecting the tip of the nose (Pn) and the most anterior point of the soft tissue chin (Pog') is defined as the *esthetic line* (E-line). It is a basic reference line for evaluating facial balance. The lower lip in white individuals should fall approximately 1 mm behind the E-line.

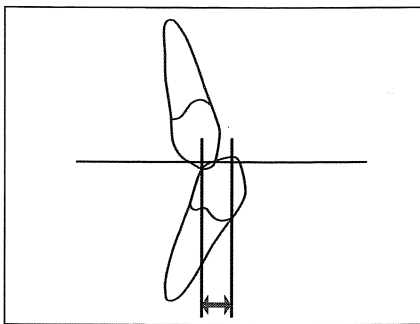


Fig 8-12 Canine relation: Describes the occlusion of the canines. Measured from the tips of the maxillary and mandibular canines along the OP.

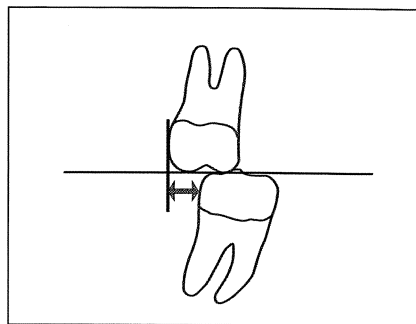


Fig 8-13 Molar relation: Describes Angle classification of occlusion. Measured from the distal surface of the mandibular molar to the distal surface of the maxillary molar along the OP.

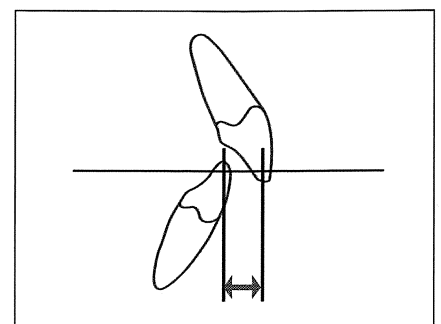


Fig 8-14 Incisor overjet: Describes the relative horizontal position of the anterior teeth. Measured from the tip of the mandibular incisor to the tip of the maxillary incisor along the OP.

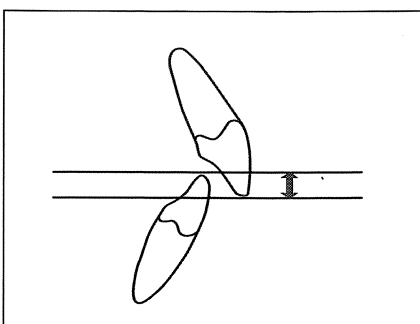


Fig 8-15 Incisor overbite: Describes the extent of the overbite. Measured from the tips of the maxillary and mandibular incisors perpendicular to the OP.

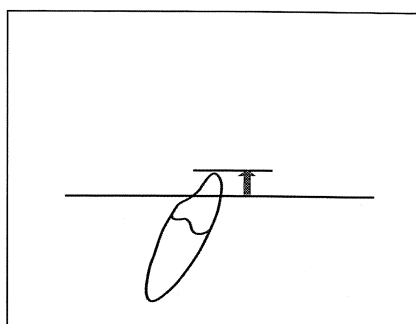


Fig 8-16 Mandibular incisor extrusion: Used to describe the cause of an overbite. Measured from the tip of the mandibular incisor to the OP.

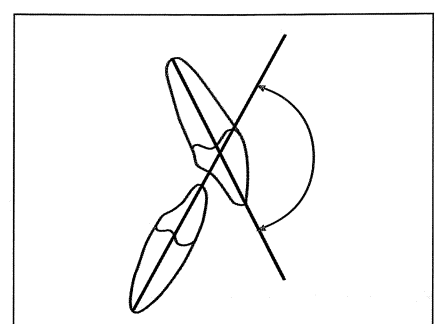


Fig 8-17 Interincisal angle: Describes both the vertical and horizontal inclination (angle) of the maxillary and mandibular incisors.

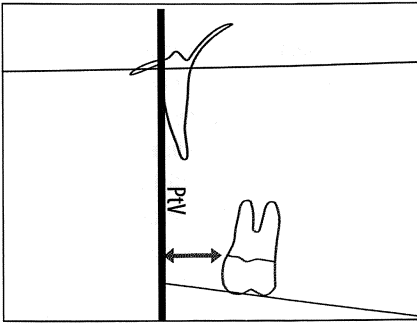


Fig 8-18 *A6 molar position to PtV*: Describes the horizontal position of the maxillary first molar. Used to determine the position of the maxillary first molar. Measured from the distal surface of the maxillary first molar to PtV.

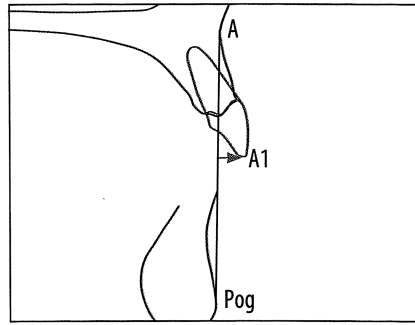


Fig 8-19 *A1 to A-Pog plane*: Describes the protrusion of the maxillary dentition. Measured from the tip of the maxillary incisor to a plane from hard tissue point A to Pog.

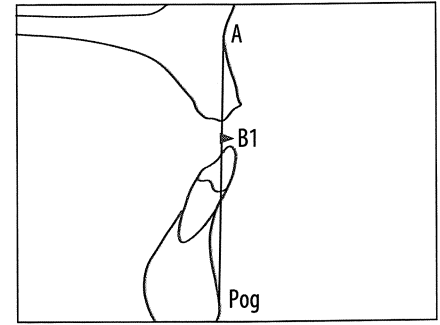


Fig 8-20 *B1 to A-Pog plane*: Describes the protrusion of the mandibular dentition. Measured from the tip of the mandibular incisor to a plane from hard tissue point A to Pog. Norm = 1.0 ± 2.5 mm. (Note: The norm listed for this measurement is considered ideal. It depends on facial pattern, stability, and personal preference of the clinician and patient.)

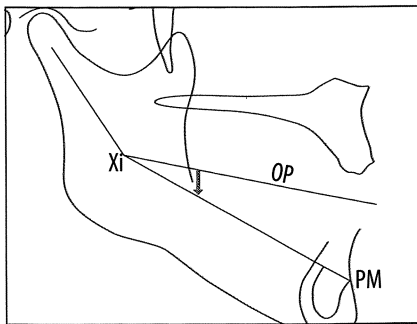


Fig 8-21 *OP to Xi*: Describes the inclination of the OP relative to the mandible. Measures the distance of the OP to the geometric center of the ramus. Also measures the angle formed by the OP and the corpus axis.

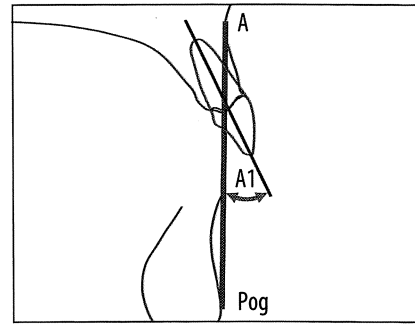


Fig 8-22 *A1 inclination to A-Pog*: Describes the position of the maxillary incisor relative to the mandible and maxilla. Measured by the angle formed by the long axis of the maxillary incisor to A-Pog plane.

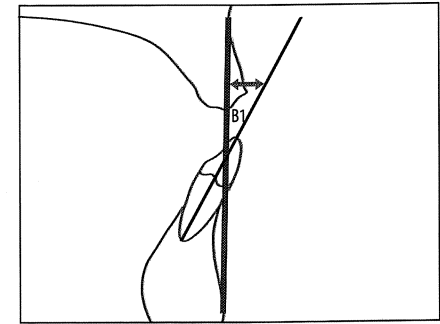


Fig 8-23 *B1 inclination to A-Pog*: Describes the position of the mandibular incisor relative to the mandible and maxilla. Measured by the angle formed by the long axis of the mandibular incisor to A-Pog plane. Norm = 22 ± 4 degrees.

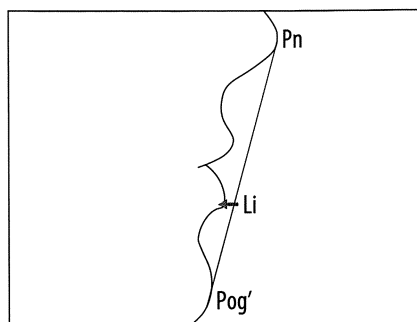
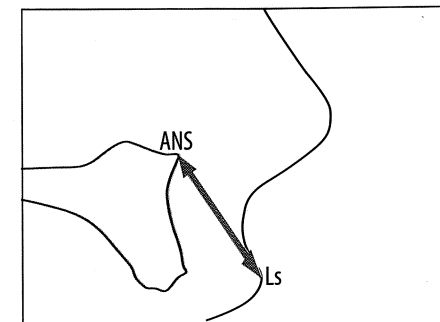


Fig 8-24 (Left) *Li to E-line*: Describes lip protrusion. Measured from the most anterior point on the lower lip to plane from Pn to Pog'.

Fig 8-25 (Right) *Upper lip length*: Measured from ANS to Ls with the lips lightly touching. Norm = 24 ± 2 mm.



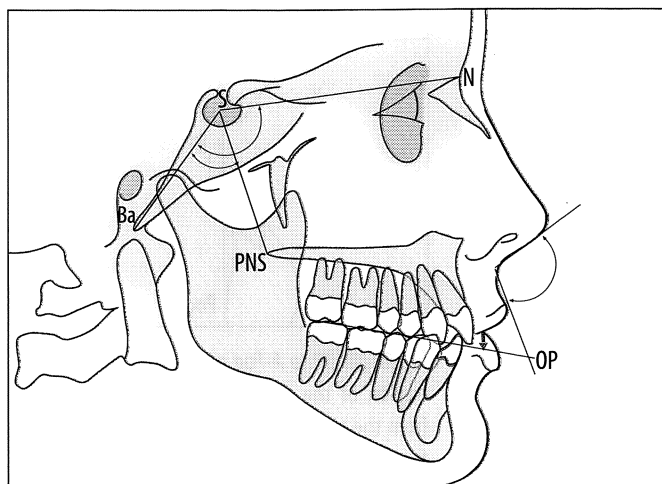


Fig 8-26 *Lip embrasure to OP*: Used to appraise soft tissue. High values may reflect a short upper lip with excessive gingival display in repose or smiling. Measured from lip embrasure to the occlusal plane. Norm = -3 mm; clinical deviation = 2 mm. *Nasolabial angle*: Measured by the angle formed by a plane from Ls to subnasale and a plane from subnasale to a tangent point on the inferior border of the nose. Norm = 115 degrees; clinical deviation = 2 mm. *Cranial base angle (NSBa)*: Used to describe the cranial base. Measured by the angle formed by the planes nasion-sella (N-S) and Ba-S. Norm = 129.6 degrees; clinical deviation = 5 degrees. *Ba-S-PNS*: Can be used to determine the horizontal position of the hard and soft palate, which is helpful for determining the cause of an airway obstruction. Measured by the angle formed by the planes Ba-S and S-PNS. Norm = 63 degrees; clinical deviation = 2.5 degrees.

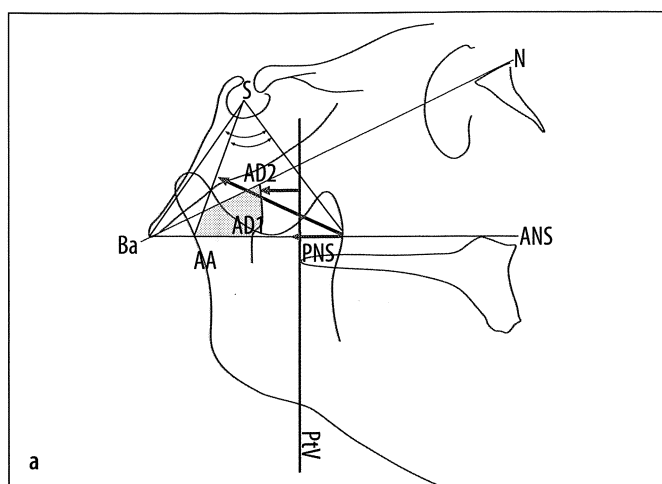


Fig 8-27a *Airway percent*: Determines the percentage of nasopharynx occupied by adenoid tissue.

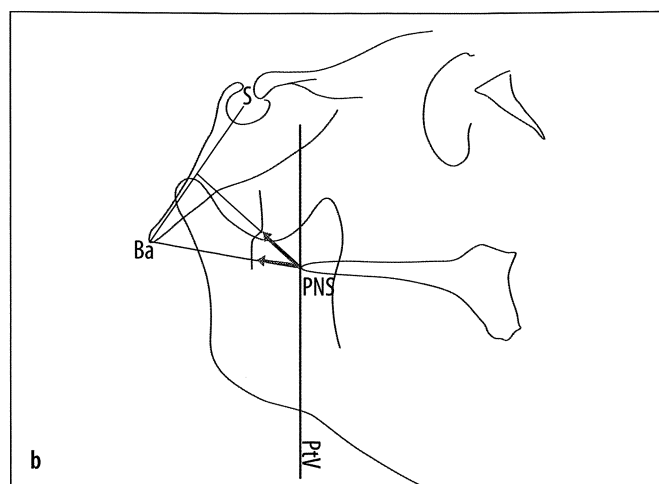


Fig 8-27b *Linder-Aronson AD1*: Used to screen for airway obstruction. Measured by the distance from PNS to the nearest adenoid tissue in a line from PNS to Ba. *Linder-Aronson AD2*: Used to screen for airway obstruction. Measured by the distance from PNS to the nearest adenoid tissue in a line from PNS perpendicular to S-Ba.

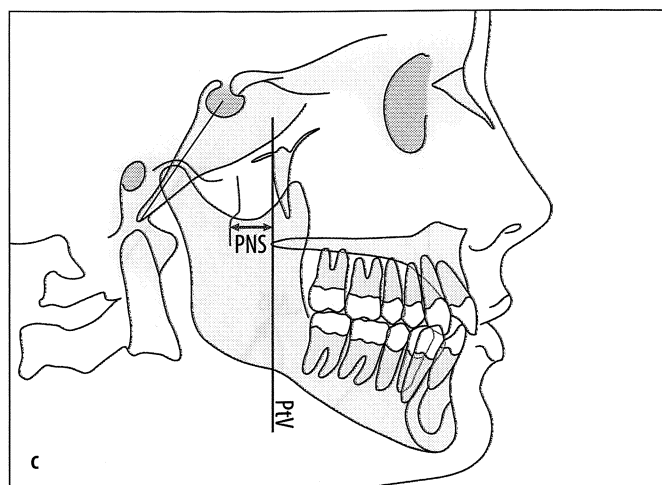


Fig 8-27c *Distance from PtV to adenoid*: Used to screen for airway obstruction. Measured from a point on PtV 5 mm superior to PNS to the nearest adenoid tissue.

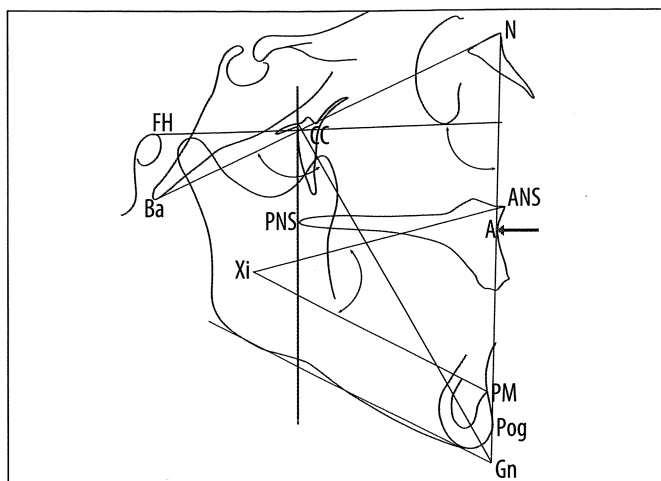


Fig 8-28 Convexity: The horizontal relationship of the maxilla to the mandible. Measured from hard tissue point A to a plane from N to Pog. Norm = 2 mm at maturity; clinical deviation = 2 mm. *Lower facial height:* The vertical relation of the mandible and maxilla. Low values indicate a skeletal deep bite. Measured by the angle formed by the planes Xi-ANS and Xi-PM. Norm = 45 degrees; clinical deviation = 4 degrees. *Facial depth:* The horizontal relationship of the mandible to the cranium. Measured by the angle formed by the planes N-Pog and FH. Norm = 86.5 degrees; clinical deviation = 3 mm. *Facial axis:* Measured by the angle formed by the planes CC-Gn and Ba-N. Norm = 90 degrees; clinical deviation = 3.5 degrees. *Maxillary depth:* The horizontal relationship of the maxilla to the cranium. Measured by the angle formed by the planes N-A and FH. Norm = 90 degrees; clinical deviation = 3 degrees.

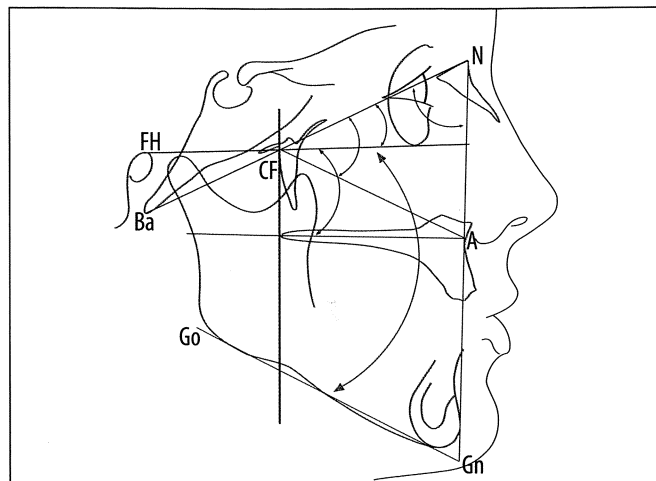


Fig 8-29 Maxillary height: The vertical relation of the maxilla to the cranium. Measured by the angle formed by the planes CF-A and CF-N. Norm = 53 ± 3 degrees. *Palatal plane to FH:* The inclination of the maxilla relative to the cranium. Measured by the angle formed by the palatal plane to FH. Norm = 1 degree; clinical deviation = 3.5 degrees. *Mandibular plane to FH:* The angle of the lower border of the mandible. Measured by the angle formed by the mandibular plane to FH. Norm = 26 degrees; clinical deviation = 4.5 degrees. *Landes angle (Ba-N-A):* The horizontal position of the maxilla to the cranium. Norm = 63 degrees; clinical deviation = 3 degrees. *Cranial deflection:* The angle formed by the planes Ba-N and FH. Norm = 27 degrees; clinical deviation = 3 degrees.

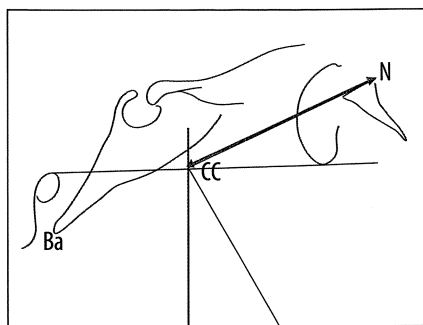


Fig 8-30 Anterior cranial length: The length of the anterior cranial base. Measured from the CC point to N along the Ba-N plane. Norm = 54.9 mm; clinical deviation = 2.5 mm.

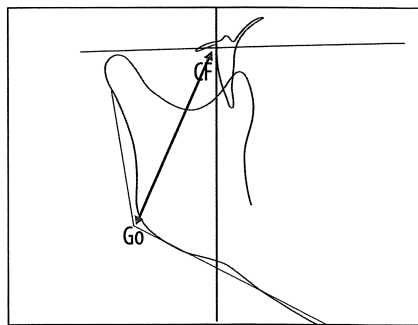


Fig 8-31 Ramus height: The height of the ramus of the mandible. Low values may indicate a more vertical facial pattern. Measured from point CF to constructed Go. Norm = 54.8 mm; clinical deviation = 3.3 mm.

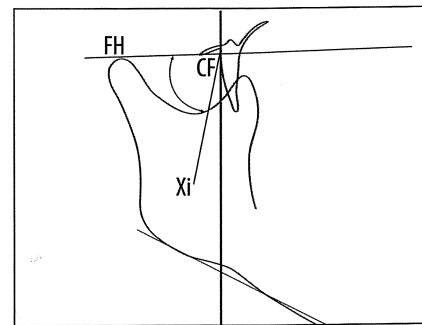


Fig 8-32 Ramus Xi position: The horizontal position of the ramus. High values may indicate abnormal mandibular growth. Measured by the angle formed by the planes CF-Xi and FH. Norm = 76 degrees; clinical deviation = 3 degrees.

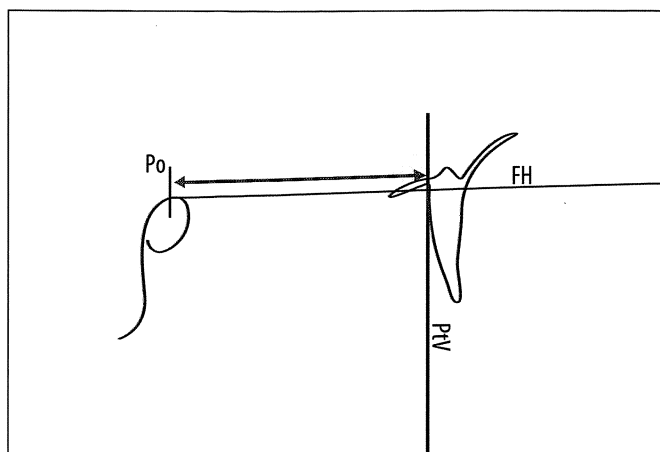


Fig 8-33 *Po location*: The anteroposterior position of Po and the glenoid fossa. Low values may indicate abnormal mandibular growth. Measured by the distance from Po to PtV along FH. Norm = -38.6 mm; clinical deviation = 2.2 mm.

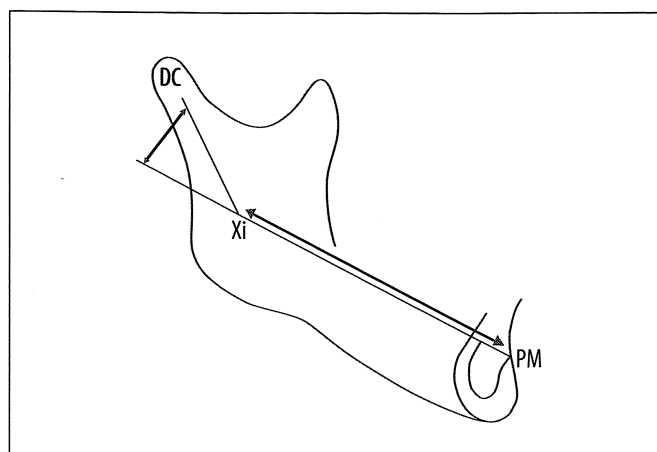


Fig 8-34 *Mandibular arc*: The angular relationship of the ramus to the body of the mandible. Measured by the angle formed by the corpus and condyle axes. Norm = 26 degrees; clinical deviation = 4 degrees.

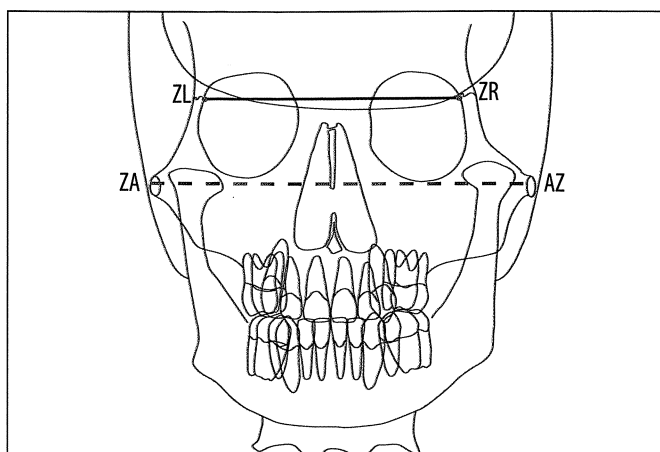


Fig 8-35 *ZL-ZR zygomatic arch*: Bilateral points on the medial margin of the zygomaticofrontal suture, at the intersection of the orbits. ZL = left; ZR = right. *ZA-AZ zygomatic arch*: Center of the zygomatic arch, midpoints. ZA = left; AZ = right.

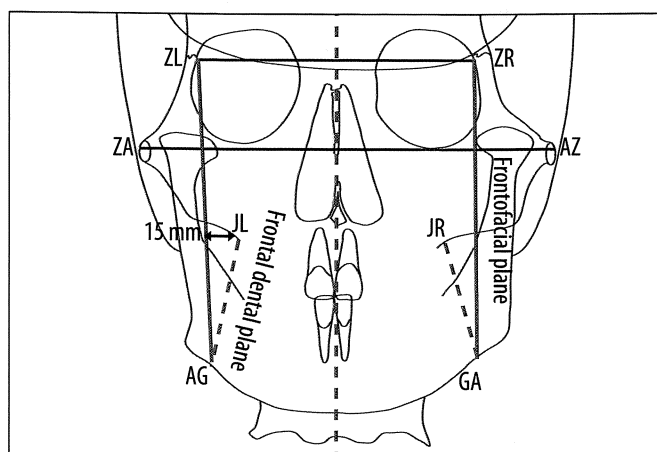


Fig 8-36 *JL/JR maxilla*: Bilateral points on the jugal process at the intersection of the outline of the tuberosity of the maxilla and zygomatic buttress. JL = left; JR = right. *Frontal dental plane*: A plane drawn from JR (and JL) to GoR (and left). *Frontofacial plane*: A plane drawn from ZR (and ZL) to the right (and left) gonial angles. The distance from point J to the frontofacial plane should be 15 mm (in an adult) and serves as a reference for locating the dentition between the dental bases. *Midsagittal plane*: A plane bisecting the head and face through the crista galli, ANS, and genial tubercles in a symmetric face.

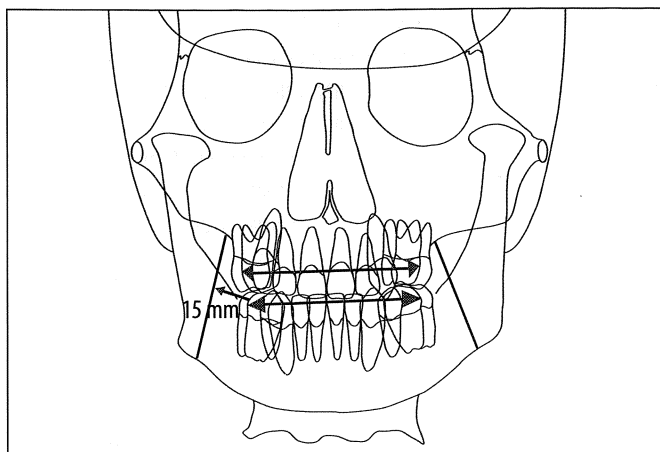


Fig 8-37 Dental plane: The distance from the buccal margin of the mandibular first molar to the dental plane is 15 mm in an adult. The distance between the buccal surfaces of the maxillary and mandibular first molars is also measured.

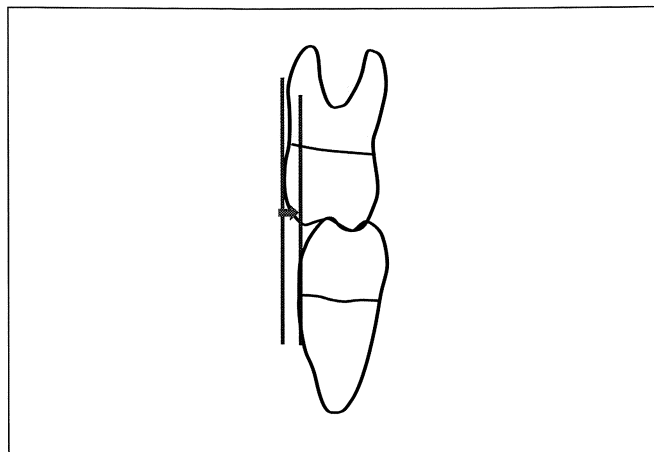


Fig 8-38 Molar relation left (A6-B6): The buccolingual occlusion of the first molars. *Molar relation right (A6-B6):* The buccolingual occlusion of the first molars. Norm = 1.5 mm; clinical deviation = 2 mm.

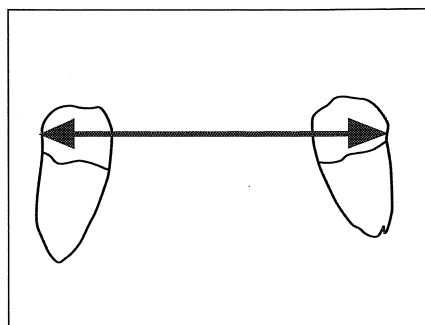


Fig 8-39a Intermolar width (B6-B6): The sagittal distance between the mandibular molars. Measured from the buccal surface of the mandibular left first molar to the buccal surface of the mandibular right first molar. Norm = 55 mm; clinical deviation = 2 mm.

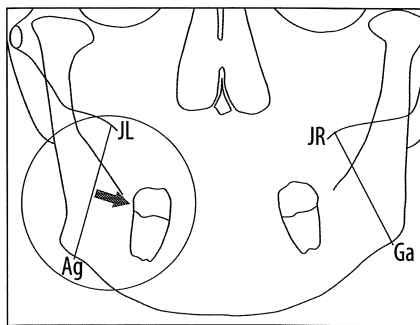


Fig 8-39b B6 to J-AB: The mandibular molar relationship to the maxilla and mandible. Measured from the buccal surface of the mandibular molar to a plane drawn from point J to the antegonial notch (Ag). Norm = 6.3 ± 1.7 mm at age 9 years.

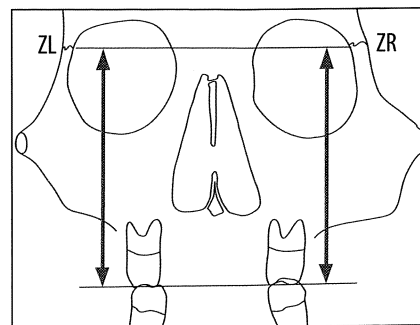


Fig 8-39c OP tilt: The difference in height between the OP and the ZL-ZR plane. Norm = 0 ± 2 mm.

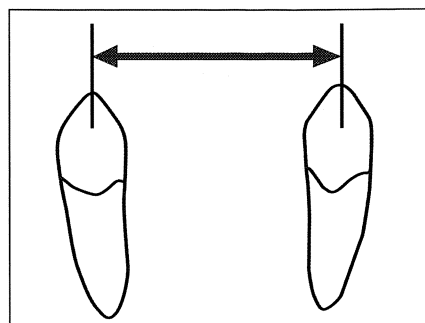


Fig 8-40 Intercanine width (B3-B3): The distance between the mandibular canines. Measured from the tip of the mandibular right canine to the tip of the mandibular left canine. Norm = 22.7 mm; clinical deviation = 2 mm.

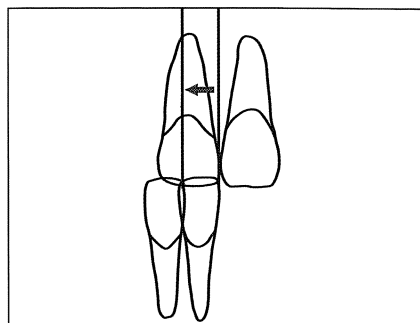


Fig 8-41a Dental midline: Used to describe a midline discrepancy. Measured from the midline of the maxillary arch to the midline of the mandibular arch. Norm = 0 mm; clinical deviation = 1.5 mm.

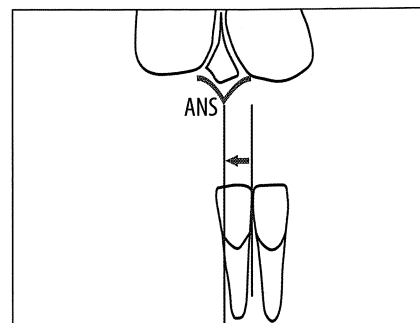


Fig 8-41b Dental midline to skeletal midline: Measured from the midline of the teeth to a line drawn from ANS to the most inferior point of the mandibular symphysis (Me).

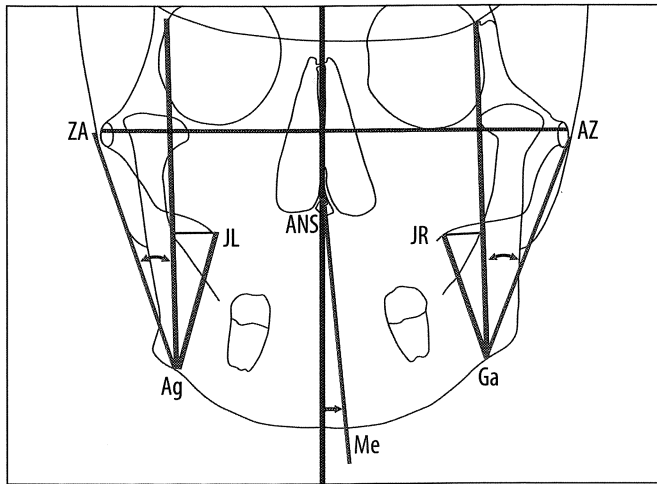


Fig 8-42 *Maxillomandibular width*: The difference between the maxilla and mandible. Measured from point J to the frontal facial plane. Norm = 11 ± 1.5 mm. *Postural symmetry*: Used to identify skeletal asymmetry. Measure of the difference in the angles (left and right) formed by two planes: zygomatic frontal suture to Ag and ZY-Ga. Norm = 0 ± 2 degrees. (Note: This measurement is affected by head positioning in the cephalostat.) *Maxillomandibular midline*: Used to describe a skeletal midline discrepancy. Measure of the angle formed by the ANS-Me plane to a plane perpendicular to ZA-AZ plane. Norm = 0 degrees; clinical deviation = 2 degrees.

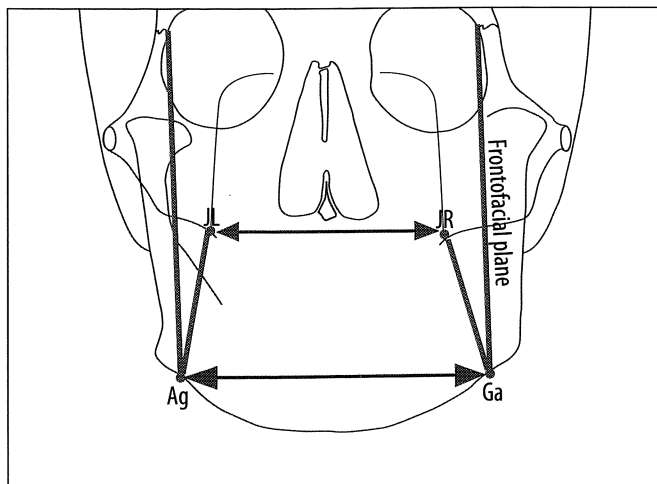


Fig 8-43 *Postural symmetry*: The difference in degrees between the left and right Ag to the left and right zygomatic arches. Norm = 0 degrees; clinical deviation = 2 degrees. (Note: This measurement is greatly affected by head positioning in the cephalostat.) *Maxillary width (JL-JR)*: The distance between the right and left J points. Norm = 61.9 mm; clinical deviation = 2 mm. *Mandibular width*: The distance between Ag and Ga along the Ag-Ga plane. Norm = 76.1 mm; clinical deviation = 2 mm.

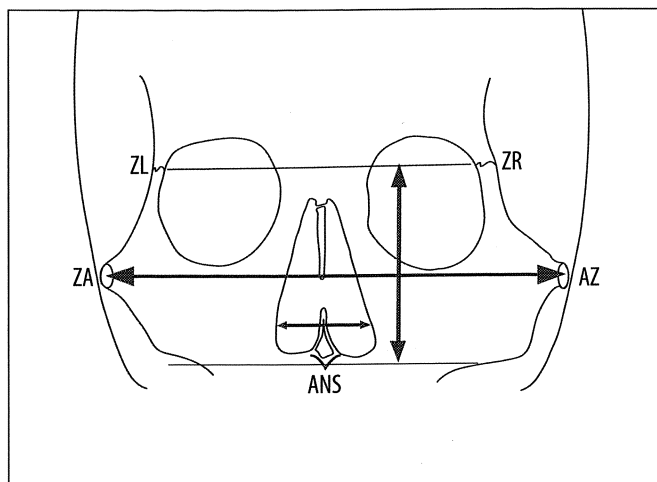


Fig 8-44 *Facial width*: The width of the face at the zygomatic arches. Measure of the distance from ZA to AZ. Norm = 115.7 mm; clinical deviation = 2 mm. *Nasal width*: The width of the nasal cavity. Measured from the widest part of the nasal cavity. Norm = 25 mm; clinical deviation = 2 mm. *Nasal height*: The height of the nasal cavity. Measure of the distance from the ZL-ZR plane to the ANS. Norm = 44.5 mm; clinical deviation = 3 mm.

Ricketts Summary Analysis				
Name of patient _____		Date _____		
Age _____		Male () Female () Traced by _____		
Chin in space _____				
	Parameter	Mean*	Adjusted mean**	Measurement
				Notable difference (Mark with asterisk*)
	(1) Facial axis	$90^\circ \pm 3.5$	No adjustment	_____
	(2) Facial (Depth) angle	$87^\circ \pm 3$	+ 1° every 3 years	_____
	(3) Mandibular plane	$26^\circ \pm 4.5$	- 1° every 3 years	_____
Convexity	(4) Convexity of point A	2 mm $\pm$ 2 mm	-1 mm every 3 years	_____
Teeth	(5) Lower incisor to A-Pog	1 mm $\pm$ 2	No adjustment	_____
	(6) Upper molar to PtV	Age + 3 mm	+ 1 mm per year	_____
	(7) Lower incisor to A-Pog	$22^\circ \pm 4$	No adjustment	_____
Profile	(8) Lower lip to E-plane	-2 mm $\pm$ 2 mm	Reduces with growth	_____
Summary of findings:				

*For 9-year-old child;				
**Adjusted to age of patient				

Fig 8-45 Ricketts Summary Analysis form, used for initial summary evaluation of the lateral cephalometric headfilm tracing.

Completing the Analysis Form

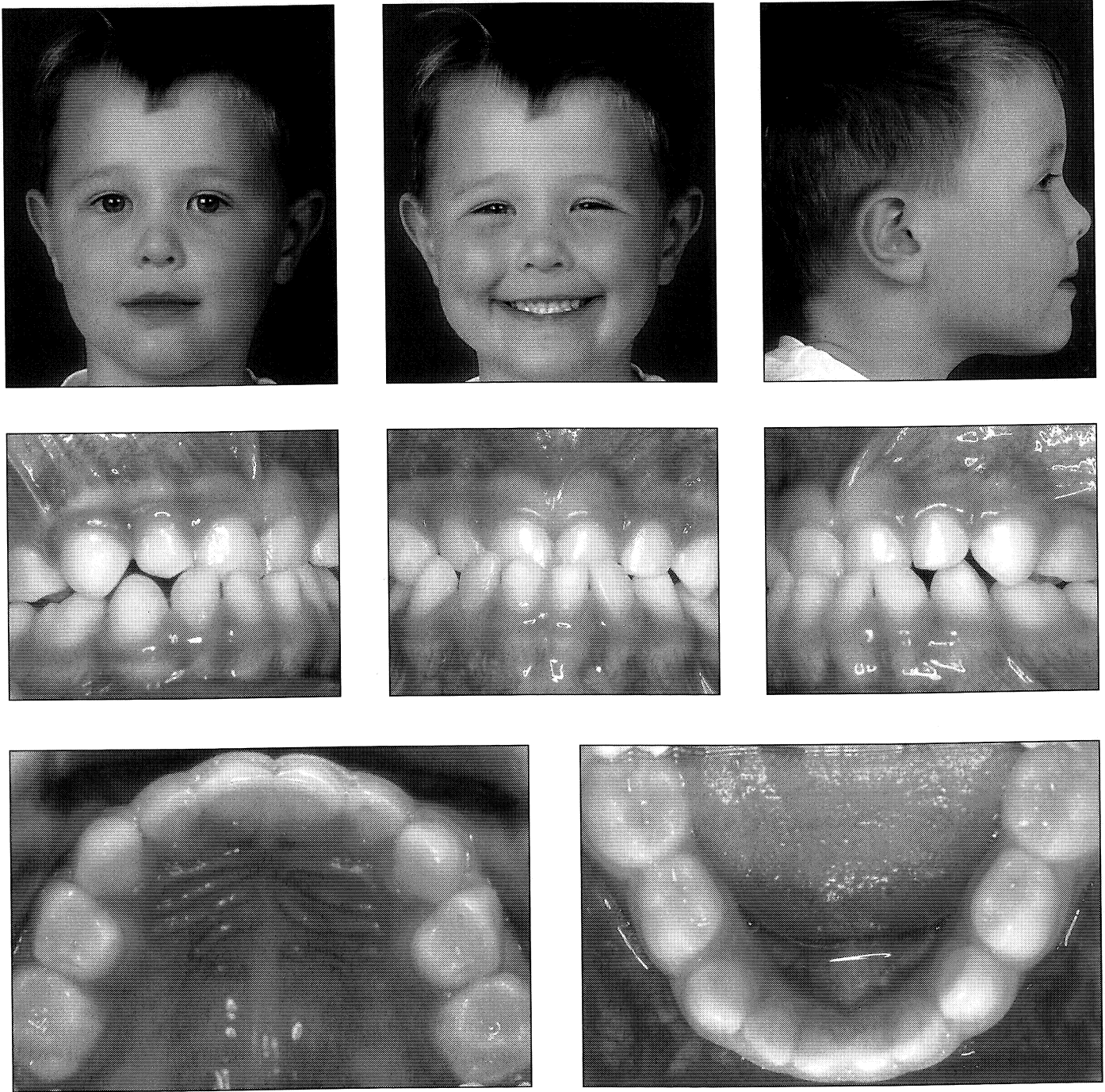
The various landmarks and planes to be analyzed on the lateral headfilm have now been identified and traced. The analysis sheet (Ricketts Summary Analysis form, Fig 8-45) lists the parameters that are to be measured and evaluated for an initial summary overview.

The first column in the table refers to "average" *mean* measurements on individuals who are 9 years of age. The second column (adjusted means) refers to the means of "average" individuals, which are adjusted to the age of the patient being evaluated.

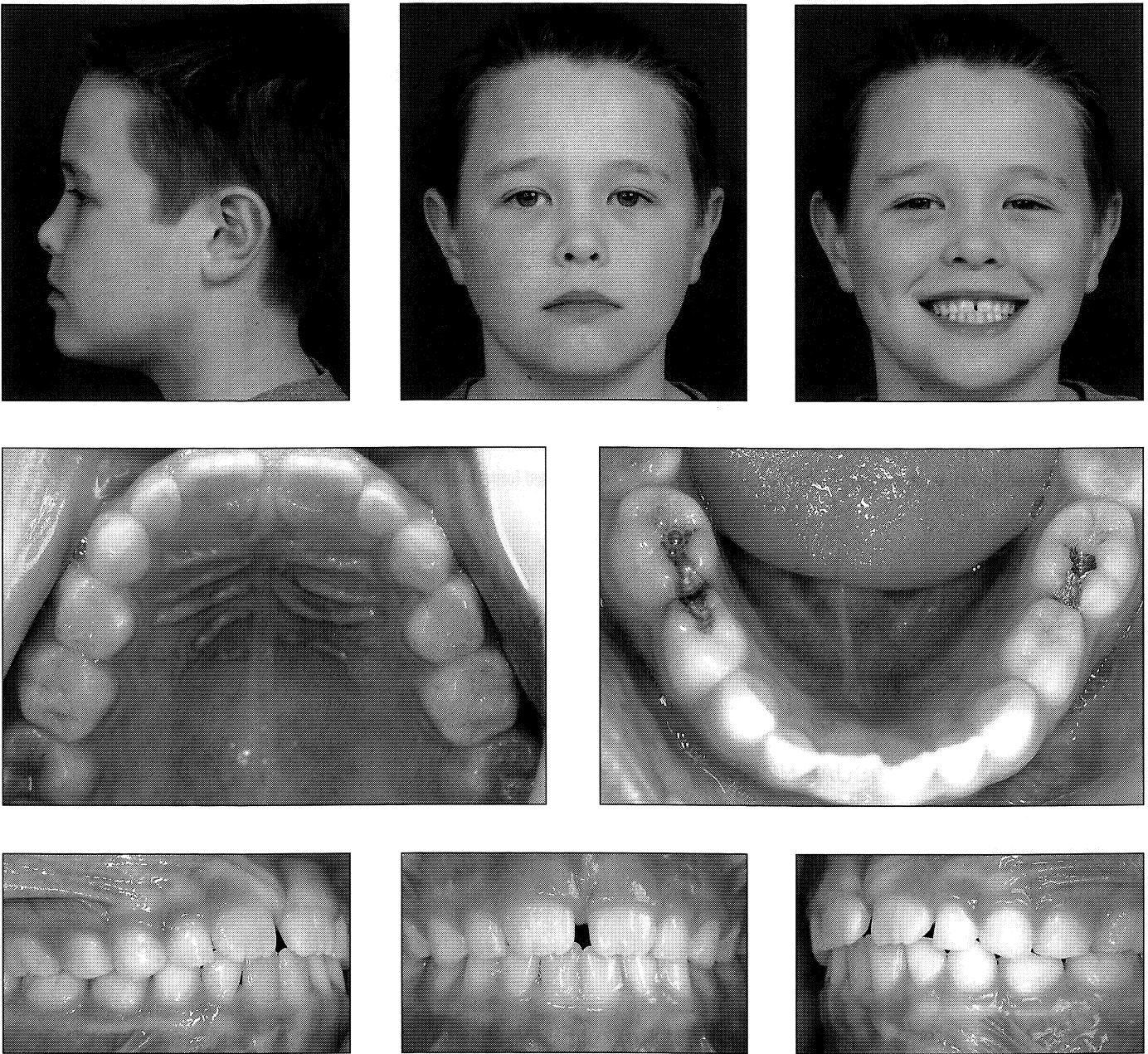
Establish the age of the patient and fill in the figures. Some figures require an adjustment; others remain unchanged. The measured parameters are now listed in the third column. For rapid and easy identification of parameters that do not approximate those of "average norms," an asterisk is placed opposite these numbers in the notable difference column. After the various discrepancies are listed, the information is correlated and a brief summary of the findings is added to the bottom of the analysis sheet.

An example of the application of the Ricketts comprehensive analysis is shown in Fig 8-46.

Fig 8-46 Application of Ricketts comprehensive analysis.



Figs 8-46a Frontal, facial, and occlusal views of a 3.9-year-old male patient with a Class III malocclusion and anterior crossbite.



Figs 8-46b Frontal, occlusal, and facial views of patient at 6.7 years following 12 months of treatment (at age 3.9 years) with a maxillary expansion appliance and reverse headgear orthopedics, resulting in improvement of the Class III skeletal malocclusion and a Class I dental occlusion.

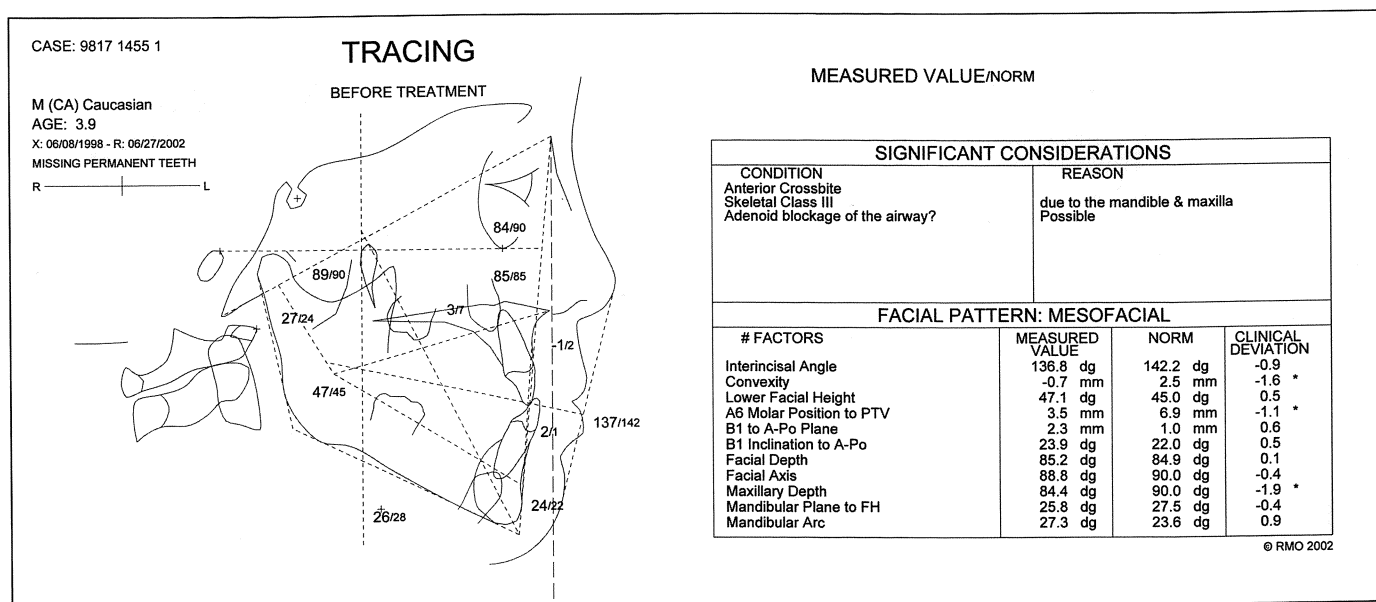


Fig 8-46c Lateral cephalometric tracing at 3.9 years. All computer-generated forms printed with permission of Rocky Mountain Orthodontics.

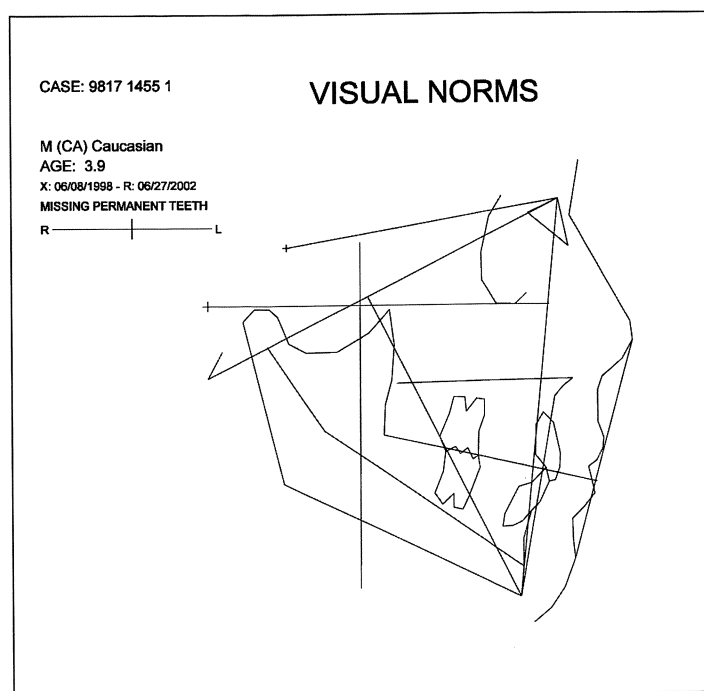


Fig 8-46d Normal cephalometric tracing template modified for age (3.9 years) and race (Caucasian), used for comparison with the patient's cephalometric tracing.

CASE: 9817 1455 1

GROWTH TO MATURITY
WITHOUT TREATMENT

RMO®

M (CA) Caucasian
AGE: 3.9
X: 05/06/1998 - R: 05/26/2002
MISSING PERMANENT TEETH

PREDICTION PERIOD to maturity

HEIGHT PREDICTION
N/A - Current height unknown

COMMENTS

ESPECIALLY PREPARED FOR DR. R. JACOBSON

GROWTH UNITS to maturity: 13.9

3RD MOLAR PREDICTION
Impaction: 69 %
Marginal: 29 %
Functional: 2 %

© RMO 2002

Fig 8-46e Ricketts emphasized the value of using a long-range growth projection until maturity to plan ahead for possible orthopedic or orthodontic interception or treatment. He emphasized the need to evaluate growing children dynamically, using these long-range projections as a guide. He also advocated the use of progress records, tracings, and growth projections for their value as visual aids during and after treatment.

CCD - ORTHODONTIC CONDITIONS									
LATERAL BEFORE TREATMENT					FRONTAL BEFORE TREATMENT				
FACTOR	MEASURED VALUE	CLINICAL NORM	CLINICAL DEVIATIONS FROM	NORM	FACTOR	MEASURED VALUE	CLINICAL NORM	CLINICAL DEVIATIONS FROM	NORM
# - Appears on tracing					# - Appears on tracing				
===== DENTAL RELATIONS =====					===== DENTAL RELATIONS =====				
01 Molar Relation	-1.5 mm	-3.0 mm	0.5		02 Molar Relation Left (A6-B6)	0.5 mm	1.5 mm	-0.5	
03 Canine Relation	-6.1 mm	-2.0 mm	-1.4 *		04 Molar Relation Right (A6-B6)	1.0 mm	1.5 mm	-0.3	
05 Incisor Overjet	2.7 mm	2.5 mm	0.1		#06 Intermolar Width (B6-B6)	57.9 mm	55.0 mm	1.4 *	
07 Incisor Overbite	0.9 mm	2.5 mm	-0.8		08 Inter canine Width (B3-B3)	22.5 mm	24.6 mm	-0.6	
09 Mand Incisor Extrusion	3.8 mm	1.3 mm	1.3 *		#10 Denture Midline	0.2 mm	0.0 mm	0.1	
#11 Interincisal Angle	129.8 dg	130.0 dg	0.0						
===== DENTAL TO SKELETON =====					===== DENTAL TO SKELETON =====				
#18 A6 Molar Position to PTV	11.7 mm	11.7 mm	0.0		19 B6 to J-Aq Left	4.7 mm	8.4 mm	-2.1 **	
#20 B1 to A-Po Plane	2.0 mm	1.0 mm	0.5		21 B6 to J-Aq Right	6.1 mm	8.4 mm	-1.3 *	
#22 A1 to A-Po Plane	4.5 mm	3.5 mm	0.5		#23 Denture to Jaw Midlines	-1.0 mm	0.0 mm	-0.6	
#24 B1 Inclination to A-Po	21.3 dg	22.0 dg	-0.2		25 Occlusal Plane Tilt	-0.5 mm	0.0 mm	-0.2	
26 A1 Inclination to A-Po	28.8 dg	28.0 dg	0.2						
27 Occlusal Plane to Xi	2.4 mm	1.0 mm	0.5						
28 Inclination of Occl Plane	18.6 dg	22.5 dg	-1.0 *						
54 B1 Inclination to FH	70.2 dg	65.0 dg	1.0 *						
===== ESTHETICS =====									
29 Lower Lip to Esthetic Plane	1.1 mm	-1.0 mm	1.1 *						
30 Upper Lip Length	26.5 mm	25.7 mm	0.4						
31 Lip Embrasure to Occl Plane	-5.4 mm	-3.0 mm	-1.2 *						
58 NasoLabial Angle	104.1 dg	115.0 dg	-2.2 **						
===== NASOPHARYNGEAL AIRWAY =====									
62 N-S-Ba	135.2 dg	129.6 dg	1.1 *						
63 Ba-S-PNS	57.8 dg	63.0 dg	-2.1 **						
85 Airway Percent	46.6 %	54.2 %	-0.5						
86 Linder-Aronson AD1	17.6 mm	21.9 mm	-0.8						
87 Linder-Aronson AD2	15.0 mm	17.2 mm	-0.6						
88 Distance PTV to Adenoid	11.3 mm	9.1 mm	0.5						

Fig 8-46f The Ricketts/RMO (Rocky Mountain Orthodontics) comprehensive lateral and frontal analyses showing the relationships between the teeth; the airway, soft tissue esthetics, and the facial structures; and the relationships between the dental and skeletal structures.

C C D - S K E L E T A L C O N D I T I O N S				
L A T E R A L B E F O R E T R E A T M E N T				
FACTOR	MEASURED VALUE	CLINICAL NORM	CLINICAL DEVIATIONS FROM NORM	
# - Appears on tracing				
===== SKELETAL RELATIONS =====				
#13 Convexity	-1.4 mm	2.1 mm	-1.7 *	
#15 Lower Facial Height	46.9 dg	45.0 dg	0.5	
#4 Present Patient Height	54.0 in	52.4 in		
91 Posterior face height	68.3 mm			
92 Anterior face height	114.1 mm			
93 Posterior/Anterior ratio	59.9 %			
94 Saddle Angle	125.4 dg	123.0 dg	0.8	
96 Condylion-A point	82.7 mm	84.4 mm	-0.7	
97 Condylion-Gnathion	113.3 mm	104.1 mm	2.2 **	
95 Max-Mand Differential	30.6 mm	19.7 mm	4.2 ***	
98 Menton-ANS	64.3 mm	62.7 mm	0.5	
===== JAW TO CRANIUM =====				
#32 Facial Depth	89.9 dg	86.5 dg	1.1 *	
#34 Facial Axis	89.5 dg	90.0 dg	-0.1	
#36 Maxillary Depth	88.5 dg	90.0 dg	-0.5	
37 Maxillary Height	57.4 dg	53.0 dg	1.5 *	
38 Palatal Plane to FH	5.9 dg	1.0 dg	1.4 *	
#39 Mandibular Plane to FH	23.8 dg	26.1 dg	-0.5	
77 Ba-N-A	60.8 dg	63.0 dg	-0.7	
76 S-N-A	74.3 dg	82.0 dg	-2.4 **	
78 S-N-B	74.6 dg	80.0 dg	-1.4 *	
69 A-N-B Difference	-0.3 dg	2.0 dg	-0.9	
75 Total Facial Height	59.2 dg	60.0 dg	-0.3	
===== INTERNAL STRUCTURE =====				
40 Cranial Deflection	30.3 dg	27.0 dg	1.1 *	
42 Cranial Length Anterior	58.7 mm	59.9 mm	-0.5	
44 Ramus Height (CF-Go)	58.0 mm	60.3 mm	-0.7	
46 Ramus Xi Position	77.6 dg	76.0 dg	0.5	
48 Porion Location (Por to PTV)	-41.2 mm	-41.7 mm	0.2	
#50 Mandibular Arc	29.3 dg	26.0 dg	0.8	
51 Corpus Length	67.9 mm	71.1 mm	-1.2 *	
FRONTAL BEFORE TREATMENT				
FACTOR	MEASURED VALUE	CLINICAL NORM	CLINICAL DEVIATIONS FROM NORM	
# - Appears on tracing				
===== SKELETAL RELATIONS =====				
#14 Max-Mand Width Left	-8.8 mm	-11.7 mm	1.9 *	
#16 Max-Mand Width Right	-8.9 mm	-11.7 mm	1.9 *	
17 Max-Mand Midline	0.6 dg	0.0 dg	0.3	
===== JAW TO CRANIUM =====				
33 Postural Symmetry	-1.3 dg	0.0 dg	-0.6	
#45 Maxillary Width (J-J)	63.0 mm	63.6 mm	-0.3	
#47 Mandibular Width (AG-GA)	76.1 mm	80.2 mm	-2.1 **	
===== INTERNAL STRUCTURE =====				
#41 Nasal Width	26.4 mm	25.0 mm	0.7	
43 Nasal Height	49.9 mm	47.4 mm	0.8	
49 Facial Width	123.8 mm	122.7 mm	0.6	

Fig 8-46g The Ricketts/RMO (Rocky Mountain Orthodontics) comprehensive lateral and frontal analyses showing the relationships between the skeletal structures; the jaw and cranial structures; and the relationships between internal structures.

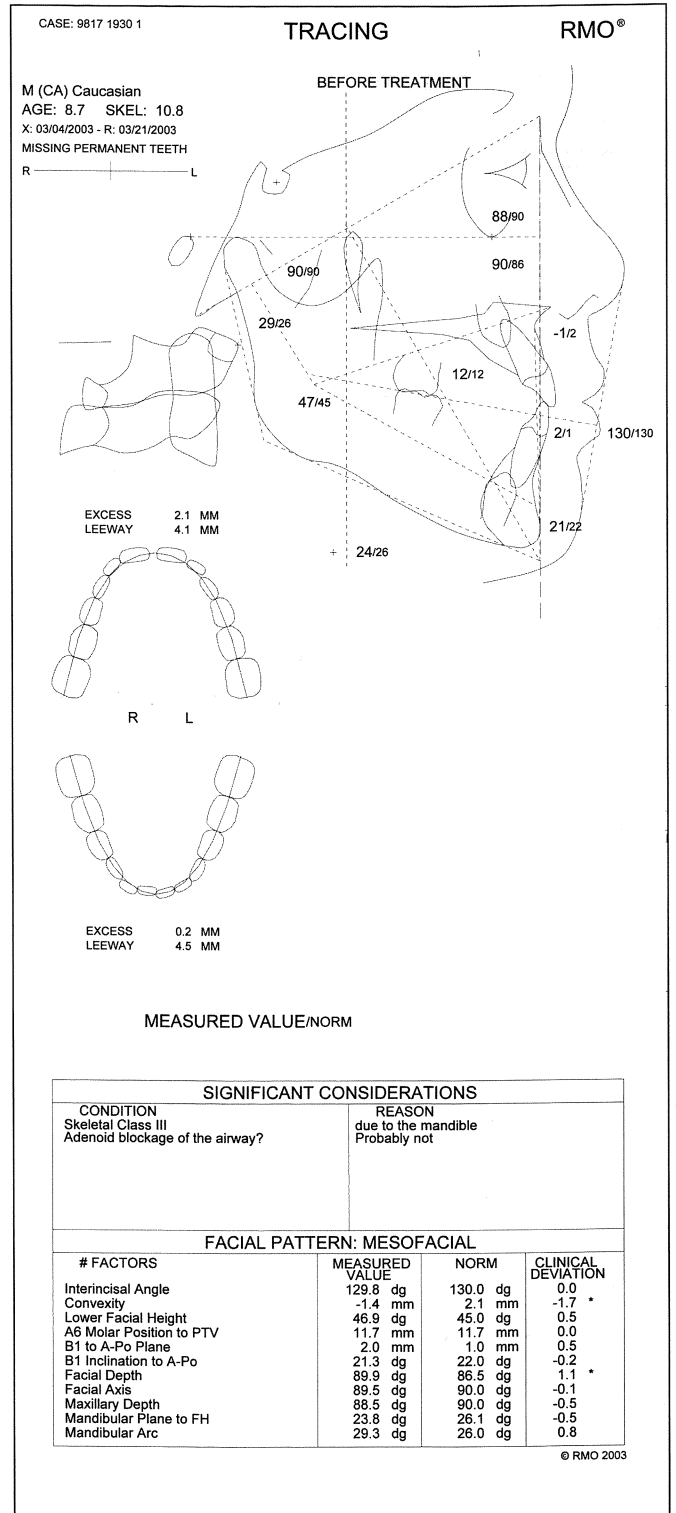


Fig 8-46h Ricketts/RMO computerized analysis and lateral and frontal tracings of the patient at 8.7 years, demonstrating a mild Class III tendency and Class I occlusion.

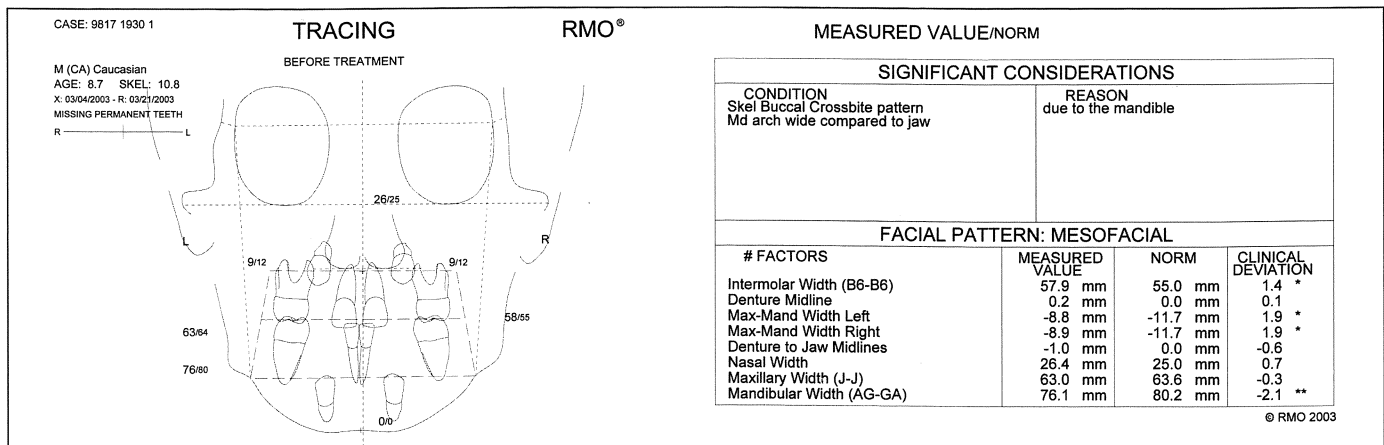
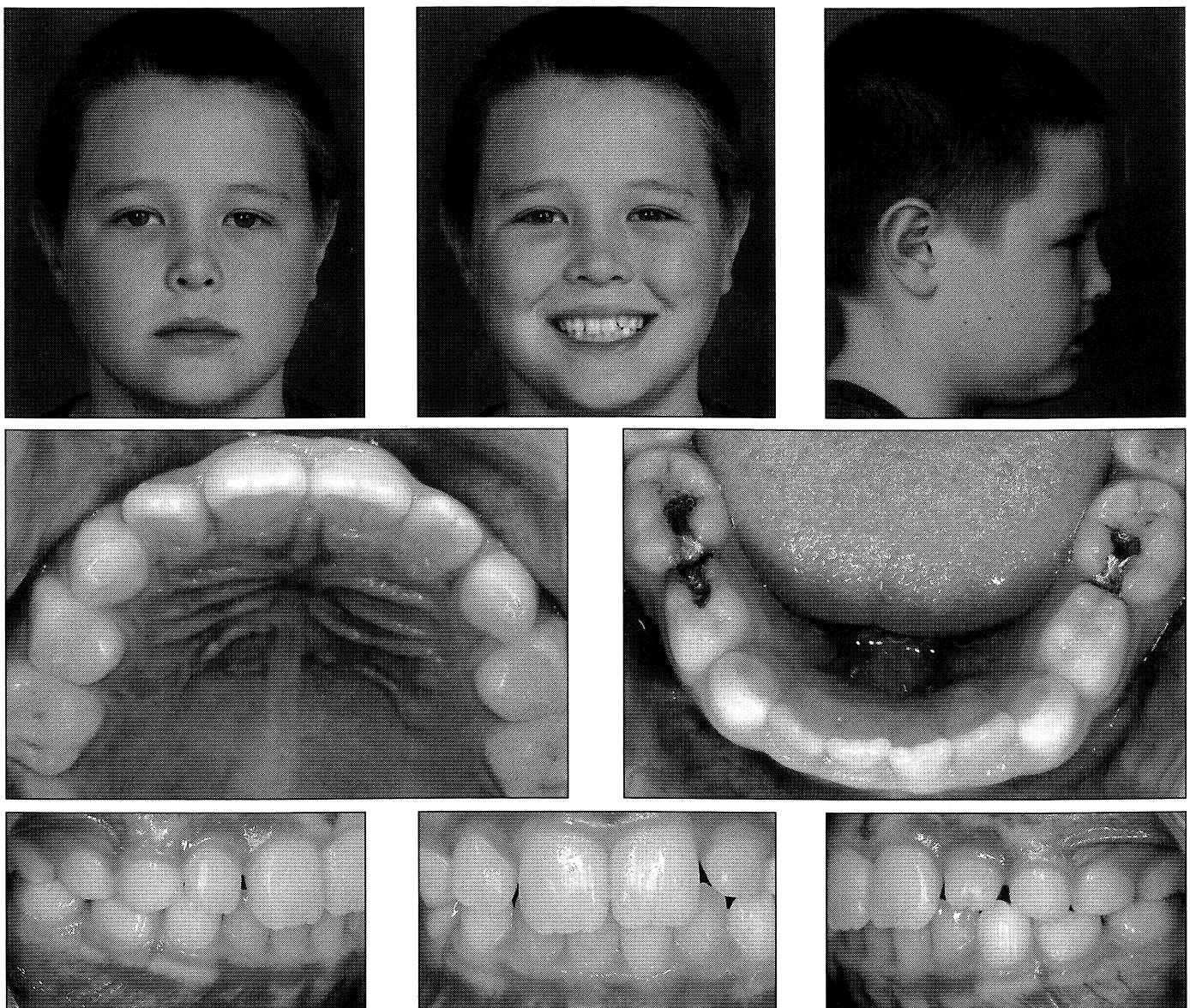


Fig 8-46i Ricketts/RMO computerized analysis and lateral and frontal tracings of the patient at 8.7 years, demonstrating a mild Class III tendency and Class I occlusion.



Figs 8-46j Clinical photographs of the patient at 10.5 years of age with no further treatment, awaiting the second phase of treatment.

Suggested Reading

- Jacobson A, Sadowsky PL. A visualized treatment objective. *J Clin Orthod* 1980;14:554-571.
- Ricketts RM. Perspectives in the clinical application of cephalometrics. *Angle Orthod* 1981;51:115-150.
- Ricketts RM. The evolution of diagnosis to computerized cephalometrics. *Am J Orthod* 1969;55:795-803.

- Ricketts RM. Clinical research in orthodontics. In: Kraus BS, Riedel RA (eds). *Vistas in Orthodontics*. Philadelphia: Lea & Febiger, 1962.
- Ricketts RM. Planning treatment on the basis of the facial pattern and an estimate of its growth. *Angle Orthod* 1957;27:14.
- Ricketts RM, Roth RH, Chaconas SJ, Schulhof RJ, Engel A. *Orthodontic Diagnosis and Planning*, Vols 1 and 2. Denver: Rocky Mountain Orthodontics, 1982.